



OctuKheir
Academy

Pharmaceutical Organic Chemistry II

Content:

Ethers

- Nomenclature
- preparation of alcohols
- Reactions of alcohols

For courses & materials:

 01067043749

 @OctuKheir



Lecture: 2

Pages: 4



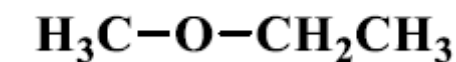
Ethers

- Ethers are compounds of formula $R-O-R'$, where R and R' may be alkyl or aryl groups.
- Ethers could be classified as;
Aliphatic ether: $(R-O-R)$,
Aryl alkyl ethers: $Ar-O-R$,
Diaryl ethers: $Ar-O-Ar$

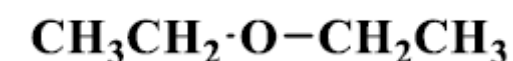
i- Nomenclature of Ethers:

A) Common name:

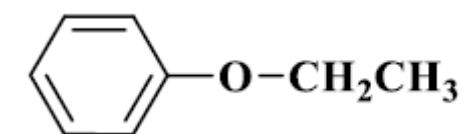
- Simple ethers with no other functional groups are named by, identifying the two Names of the two alkyl or aryl groups followed by the word ether.



Ethyl methyl ether



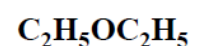
Diethyl ether



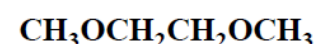
ethyl phenyl ether

B) IUPAC name:

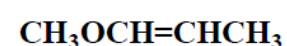
- Alkoxy group $(RO-)$, is not functional groups; it is substitution as -R & -X
- Ethers are named as; alkoxyalkanes, alkoxyalkenes, and alkoxyarenes.



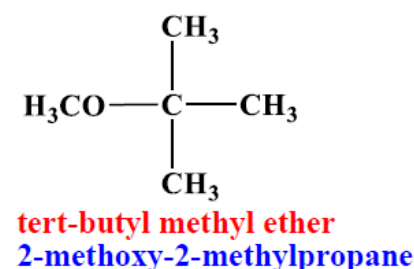
diethyl ether
ethoxyethane



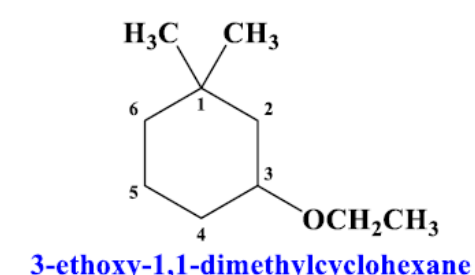
1,2-dimethoxyethane



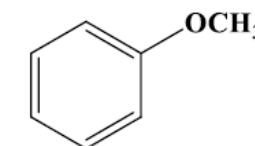
1-methoxy-1-propene



tert-butyl methyl ether
2-methoxy-2-methylpropane



3-ethoxy-1,1-dimethylcyclohexane

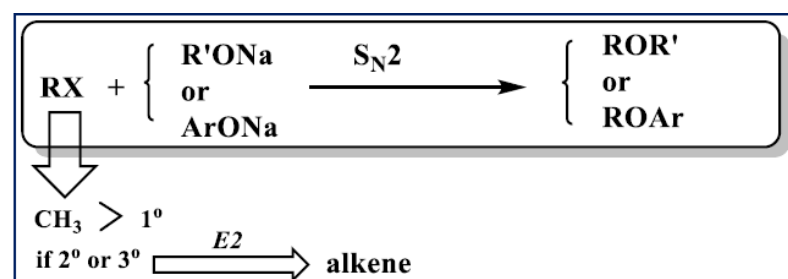


Methyl phenyl ether (anisole)
methoxybenzene

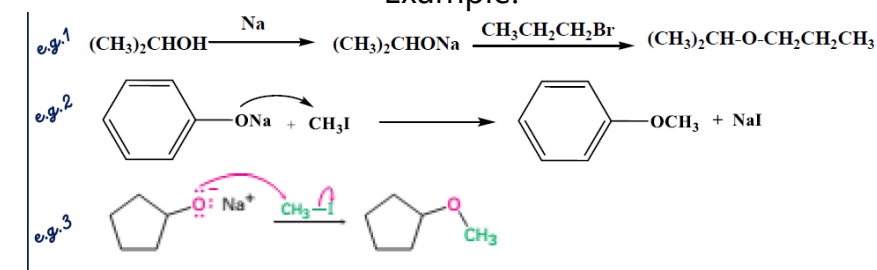
ii- Preparation of ether:

1- Williamson's synthesis:

- Through S_N2 reaction of alkyl halides RX with alkoxide $R'ONa$ or phenoxides $ArONa$

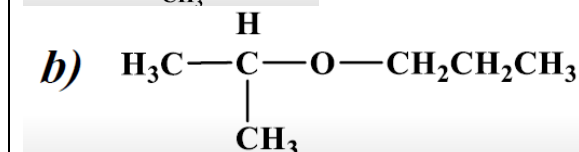
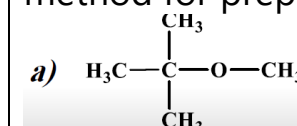


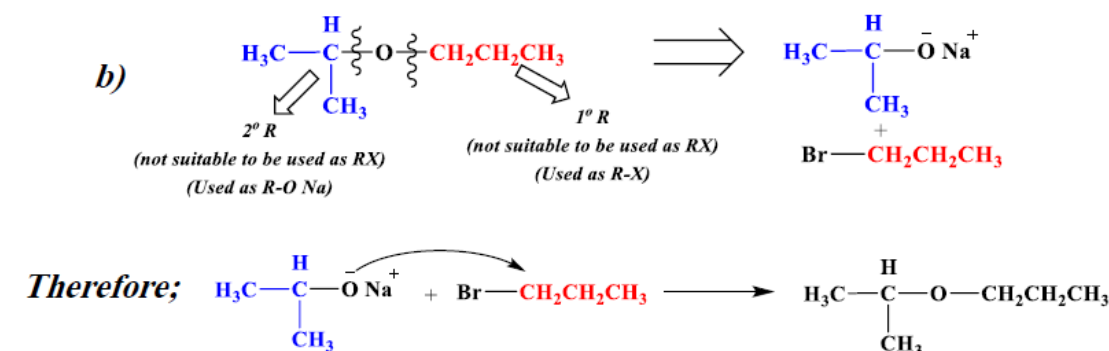
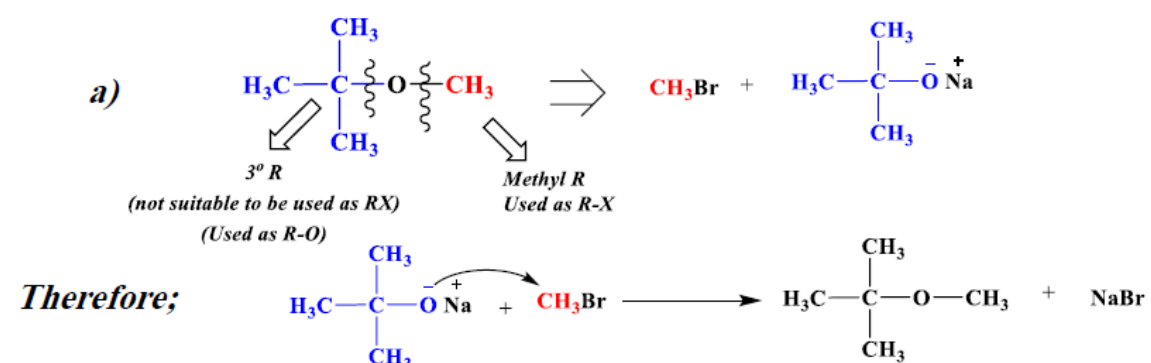
N.B. For Williamson's synthesis, RX must be Methyl or 1°
Example:



Test your self:

Q1: Prepare the following ether (Suggest the best method for preparation of the following ethers)?

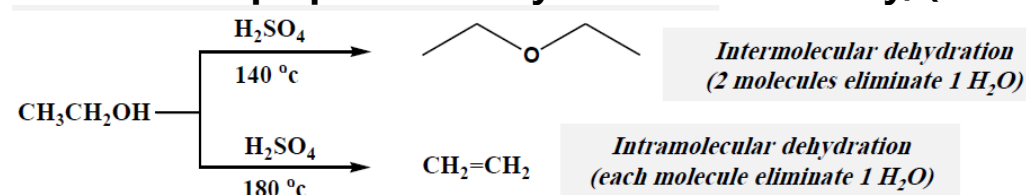




2- Intermolecular dehydration of alcohols:

N.B. the reaction is catalyzed by acids (H₂SO₄).

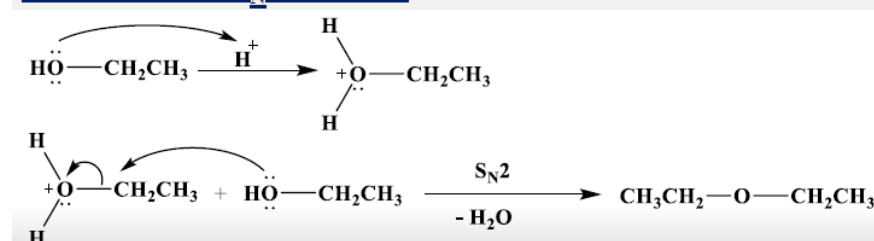
N.B. Used for preparation of symmetric ethers only, (R-O-R).



N.B. Only 1o ROH could be used to prepare ether through this method, 2o & 3o will undergo E1 reaction to give alkene (intramolecular dehydration).

N.B. for 1o ROH may undergo intermolecular dehydration to form ether or intramolecular dehydration to give alkene according to reaction temperature.

Mechanism of S_N2 reaction



iii- Physical Properties of Ether (R-O-R):

□ Boiling Point: -

• B.P. of ether lower than that of isomeric alcohols (no intermolecular H-bonding).

□ Water solubility: -

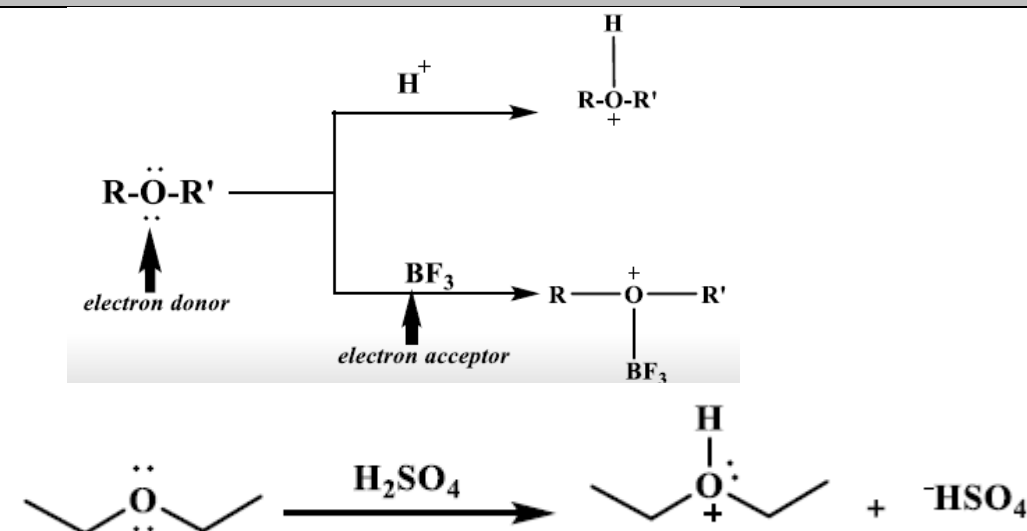
• Ethers have similar water solubility to those of isomeric alcohols (form Hbonding e\ water).

iv- Reactions of ethers (R-O-R):

□ Ether are nearly unreactive as ether linkage are stable towards bases, oxidizing agents & reducing agent.

1- Ethers react as bases: With acids & Lewis acids to form oxonium ions.

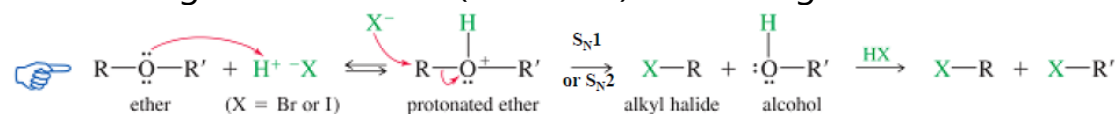
e.g.1





2-Cleavage of Ether linkage:

- Heating ethers with acids (HI or HBr) → breakage of ether bond.

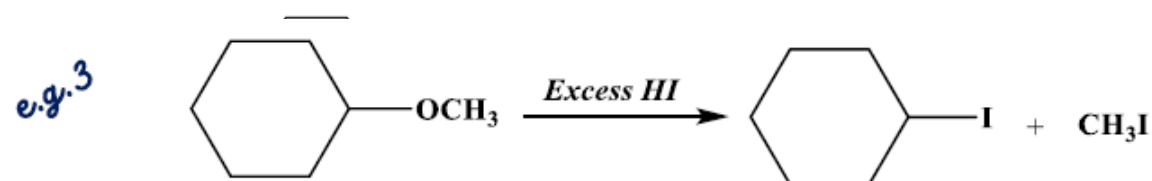
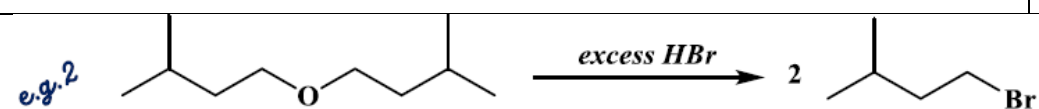
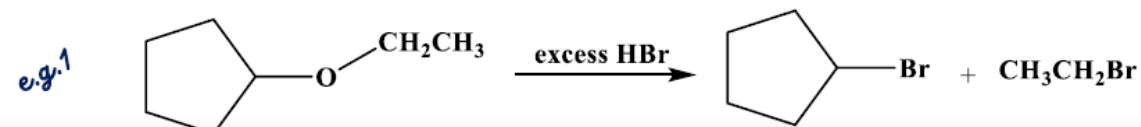


- Order of reactivity of HX: HI > HBr

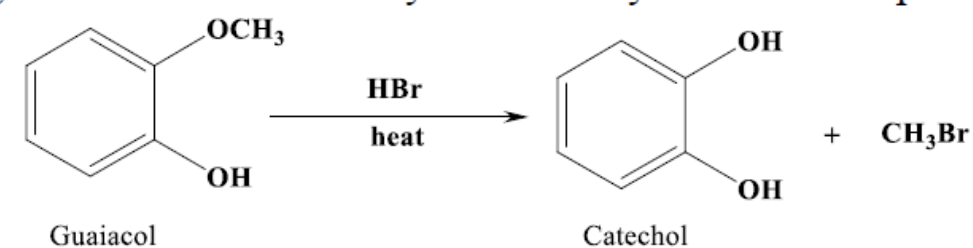
- The reaction can proceed via S_N1 & S_N2 depending on the structure of R
- 1° R reacts via S_N2 while 3° R reacts via S_N1

- The ideal acid: strong acid with strong nucleophilic counter ion e.g. HI.

Example:



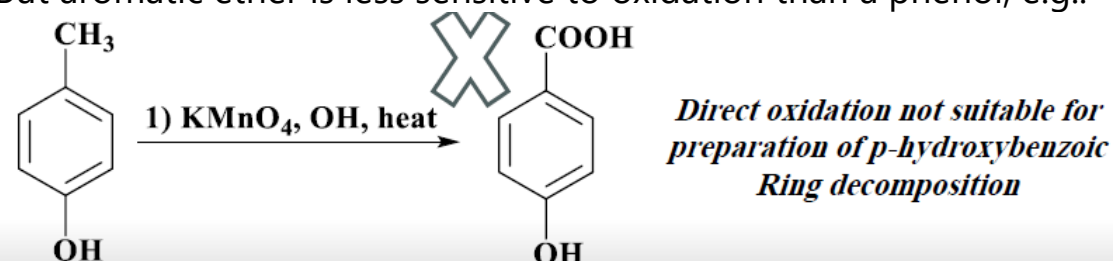
- In case aryl alkyl ether (Ar-O-R); the cleavage always proceeds so that the alkyl-oxygen bond is broken and yields an alkyl halide and a phenol.



3- Electrophilic substitution reactions of aryl ether:

- Alkoxy group (RO-) is a moderate activating group, o- & p- directing.

But aromatic ether is less sensitive to oxidation than a phenol, e.g.:



Protection of the hydroxy group (-OH) through etherification, followed by oxidation:

